35-10-1 Flight Deck Fixed Oxygen System OXYGEN Pressure Gauges

Interval	Installed	Required	Procedure
C	2	1	(M)

Either the Flight Compartment Gauge or Cylinder Mounted Gauge may be inoperative provided there is another method to confirm that adequate oxygen is available for the intended flight.

Minimum Equipment List Procedures (MELP)

Placard

Placard inoperative Flight Deck Fixed Oxygen System Pressure Gauge in the flight compartment, clearly indicating which pressure gauge is inoperative.

Operating Procedures

None required.

Maintenance Procedures

1. Place a placard on the co-pilot’s side console clearly indicating which pressure gauge is inoperative (bottle gauge or flight deck gauge).
2. If the inoperative gauge is the oxygen bottle gauge, also placard the bottle gauge in the nose compartment as inoperative.
3. Prior to each flight, check the pressure in the Flight Deck Fixed Oxygen System bottle by whichever pressure gauge is operative.
4. The minimum dispatch pressure in the 3 crew member configuration for the 39.8 cu.ft cylinder is 1800 psig at 70 °F (21°C). Refer to Figure 35-1, Figure 35-2 and the following Figure 35-3, Table 1 for pressure variations with change in ambient temperature for 39.8 cu. ft cylinder.
5. The minimum dispatch pressure in the 3 crew member configuration for the 50.0 cu.ft cylinder is 1450 psig at 70 °F (21°C). Refer to Figure 35-4, Figure 35-5 and the following Figure 35-6, Table 2 for pressure variations with change in ambient temperature for the 50.0 cu. ft cylinder.
6. Make appropriate entry in the aircraft journey log.

(…Continued)

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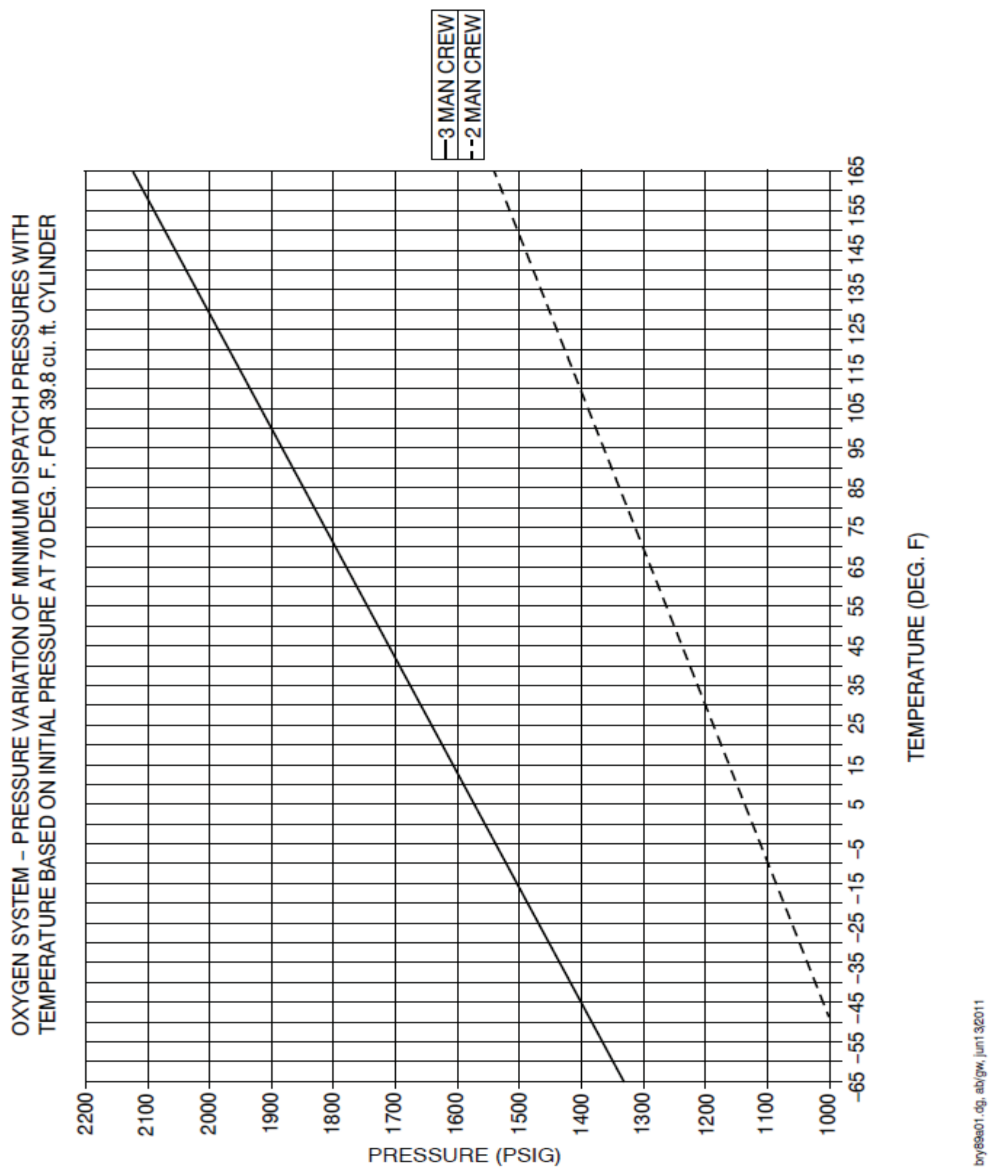
35-10-1 Flight Deck Fixed Oxygen System OXYGEN Pressure Gauges



OXYGEN SYSTEM – PRESSURE VARIATION OF MIN. DISPATCH PRESSURES WITH TEMP. BASED ON INITIAL PRESSURE AT 70 DEG. F FOR 39.8 CU. FT. CYLINDER

Figure 35–1

(…Continued)

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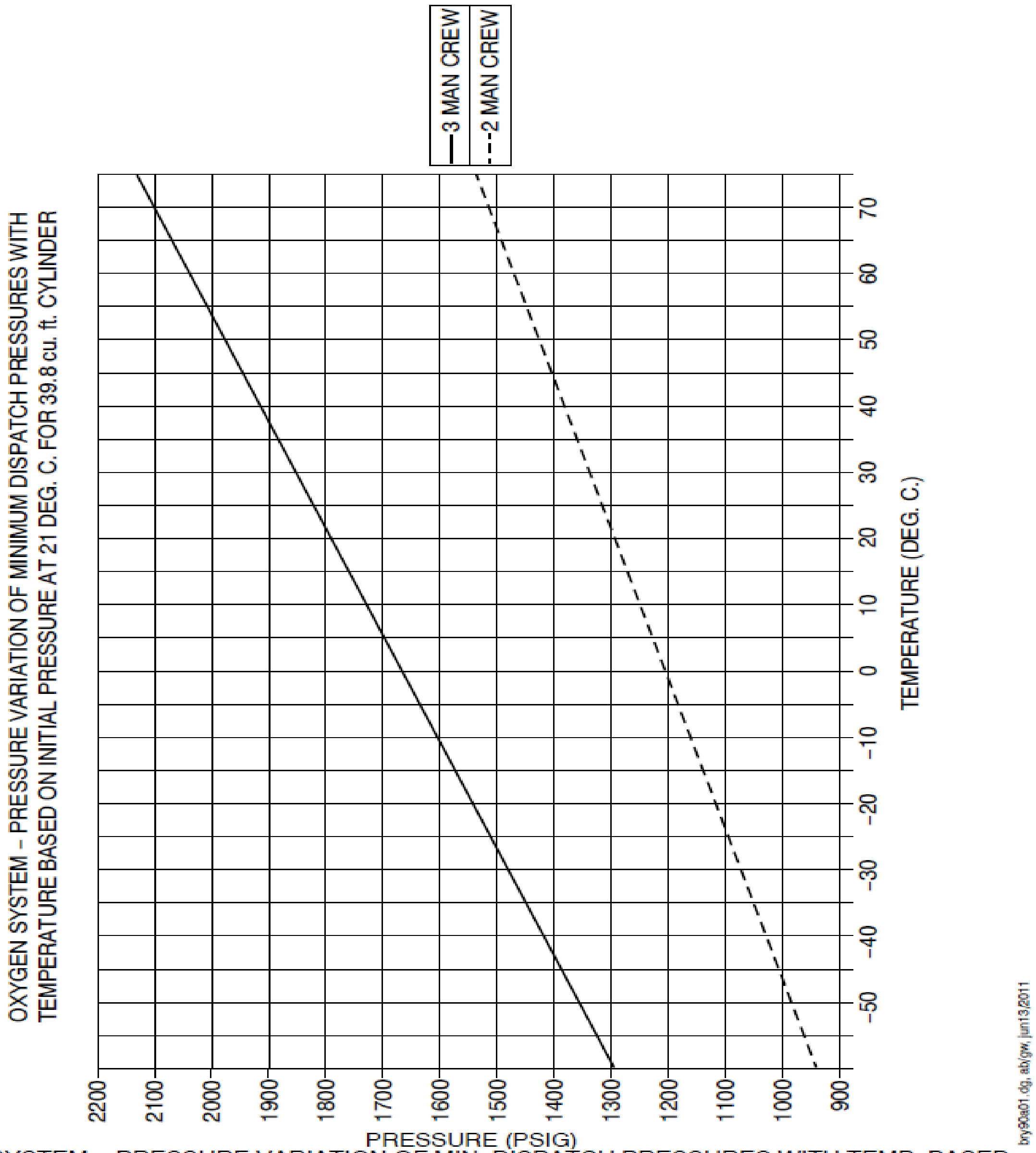
35-10-1 Flight Deck Fixed Oxygen System OXYGEN Pressure Gauges



OXYGEN SYSTEM – PRESSURE VARIATION OF MIN. DISPATCH PRESSURES WITH TEMP. BASED ON INITIAL PRESSURE AT 21 DEG. C FOR 39.8 CU. FT. CYLINDER

Figure 35–2

(…Continued)

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35-10-1 Flight Deck Fixed Oxygen System OXYGEN Pressure Gauges

PRESSURE VARIATION OF MINIMUM DISPATCH PRESSURES WITH TEMPERATURE, BASED ON INITIAL TEMPERATURE AT 70 DEG. F (21 DEG. C) FOR 39.8 cu. ft CYLINDER			
CYLINDER TEMPERATURE		MINIMUM DISPATCH PRESSURE – PSIG	
°F	°C	3 CREW	2 CREW
–65	–53.9	1341	969
–60	–51.1	1358	981
–55	–48.3	1375	993
–50	–45.6	1392	1005
–45	–42.8	1409	1018
–40	–40.0	1426	1030
–35	–37.2	1443	1042
–30	–34.4	1460	1055
–25	–31.7	1477	1067
–20	–28.9	1494	1079
–15	–26.1	1511	1091
–10	–23.3	1528	1104
–5	–20.6	1545	1116
0	–17.8	1562	1128
5	–15.0	1579	1140
10	–12.2	1596	1153
15	–9.4	1613	1165
20	–6.7	1630	1177
25	–3.9	1647	1190
30	–1.1	1664	1202
35	1.7	1681	1214
40	4.4	1698	1226
45	7.2	1715	1239
50	10.0	1732	1251
55	12.8	1749	1263
60	15.6	1766	1275
65	18.3	1783	1288
70	21.1	1800	1300
75	23.9	1817	1312
80	26.7	1834	1325
85	29.4	1851	1337
90	32.2	1868	1349

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TABLE 1 (SHEET 1 OF 2)

Figure 35–3

(…Continued)

35-10-1 Flight Deck Fixed Oxygen System OXYGEN Pressure Gauges

PRESSURE VARIATION OF MINIMUM DISPATCH PRESSURES WITH TEMPERATURE, BASED ON INITIAL TEMPERATURE AT 70 DEG. F (21 DEG. C) FOR 39.8 cu. ft CYLINDER			
CYLINDER TEMPERATURE		MINIMUM DISPATCH PRESSURE – PSIG	
°F	°C	3 CREW	2 CREW
95	35.0	1885	1361
100	37.8	1902	1374
105	40.6	1919	1386
110	43.3	1936	1398
115	46.1	1953	1410
120	48.9	1970	1423
125	51.7	1987	1435
130	54.4	2004	1447
135	57.2	2021	1460
140	60.0	2038	1472
145	62.8	2055	1484
150	65.6	2072	1496
155	68.3	2089	1509
160	71.1	2106	1521
165	73.9	2123	1533

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TABLE 1 (SHEET 2 OF 2)

Figure 35–3

(…Continued)

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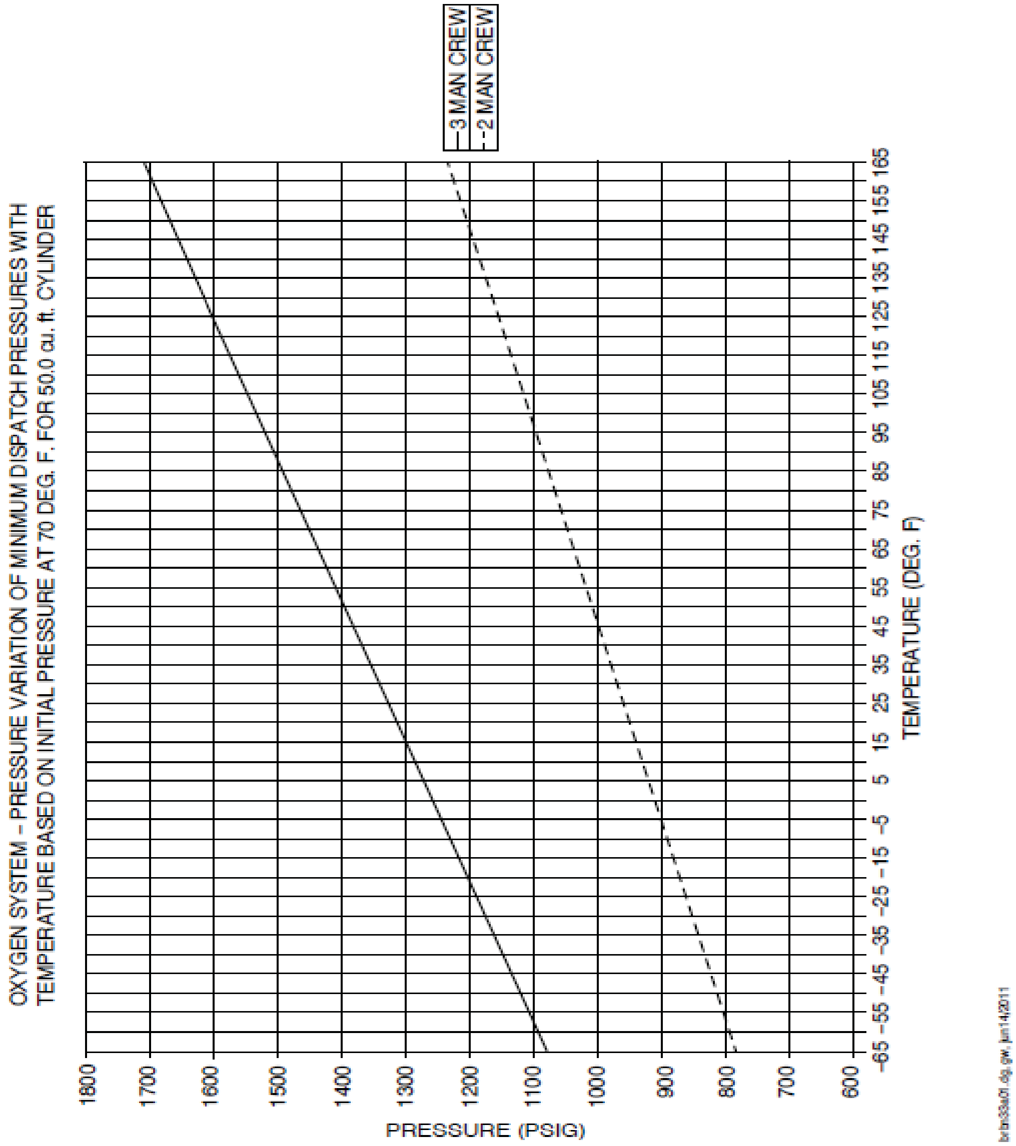
35-10-1 Flight Deck Fixed Oxygen System OXYGEN Pressure Gauges



OXYGEN SYSTEM – PRESSURE VARIATION OF MIN. DISPATCH PRESSURES WITH TEMP. BASED ON INITIAL PRESSURE AT 70 DEG. F FOR 50.0 CU. FT. CYLINDER
Figure 35–4

(…Continued)

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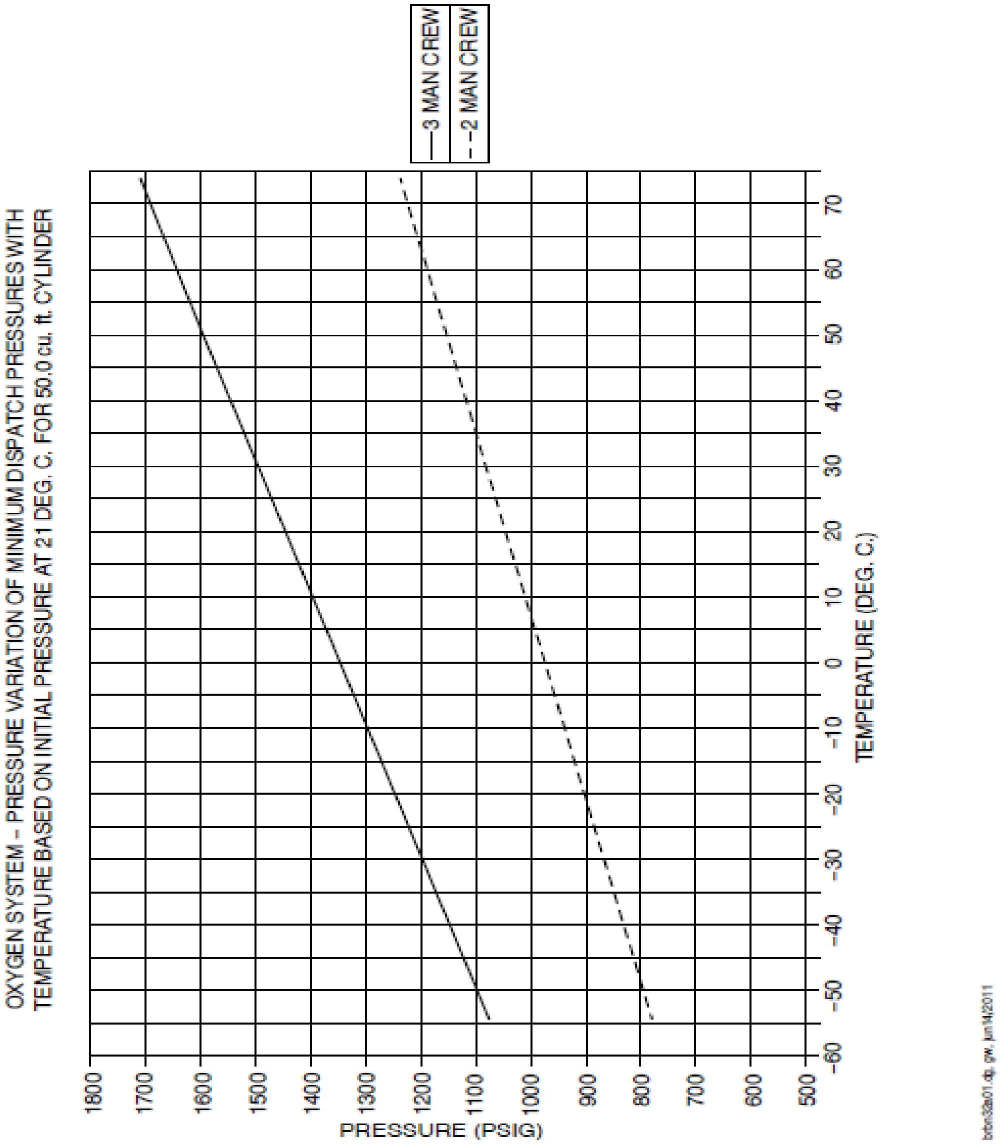
35-10-1 Flight Deck Fixed Oxygen System OXYGEN Pressure Gauges



OXYGEN SYSTEM – PRESSURE VARIATION OF MIN. DISPATCH PRESSURES WITH TEMP. BASED ON INITIAL PRESSURE AT 21 DEG. C FOR 50.0 CU. FT. CYLINDER
Figure 35–5

(…Continued)

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35-10-1 Flight Deck Fixed Oxygen System OXYGEN Pressure Gauges

PRESSURE VARIATION OF MINIMUM DISPATCH PRESSURES WITH TEMPERATURE, BASED ON INITIAL TEMPERATURE AT 70 DEG. F (21 DEG. C) FOR 50.0 cu. ft CYLINDER			
CYLINDER TEMPERATURE		MINIMUM DISPATCH PRESSURE – PSIG	
°F	°C	3 CREW	2 CREW
–65	–53.9	1080	782
–60	–51.1	1094	792
–55	–48.3	1108	802
–50	–45.6	1121	812
–45	–42.8	1135	822
–40	–40.0	1149	832
–35	–37.2	1163	842
–30	–34.4	1176	852
–25	–31.7	1190	862
–20	–28.9	1204	872
–15	–26.1	1217	881
–10	–23.3	1231	891
–5	–20.6	1245	901
0	–17.8	1258	911
5	–15.0	1272	921
10	–12.2	1286	931
15	–9.4	1299	941
20	–6.7	1313	951
25	–3.9	1327	961
30	–1.1	1340	971
35	1.7	1354	981
40	4.4	1368	991
45	7.2	1382	1000
50	10.0	1395	1010
55	12.8	1409	1020
60	15.6	1423	1030
65	18.3	1436	1040
70	21.1	1450	1050
75	23.9	1464	1060
80	26.7	1477	1070
85	29.4	1491	1080
90	32.2	1505	1090

TABLE 2 (SHEET 1 OF 2)

Figure 35–6

(…Continued)

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35-10-1 Flight Deck Fixed Oxygen System OXYGEN Pressure Gauges

PRESSURE VARIATION OF MINIMUM DISPATCH PRESSURES WITH TEMPERATURE, BASED ON INITIAL TEMPERATURE AT 70 DEG. F (21 DEG. C) FOR 50.0 cu. ft CYLINDER			
CYLINDER TEMPERATURE		MINIMUM DISPATCH PRESSURE – PSIG	
°F	°C	3 CREW	2 CREW
95	35.0	1518	1100
100	37.8	1532	1109
105	40.6	1546	1119
110	43.3	1560	1129
115	46.1	1573	1139
120	48.9	1587	1149
125	51.7	1601	1159
130	54.4	1614	1169
135	57.2	1628	1179
140	60.0	1642	1189
145	62.8	1655	1199
150	65.6	1669	1209
155	68.3	1683	1219
160	71.1	1696	1228
165	73.9	1710	1238

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TABLE 2 (SHEET 2 OF 2)

Figure 35–6

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35-10-2 Flight Deck Fixed Oxygen System Overboard Discharge Indicator

Interval	Installed	Required	Procedure
C	1	0	(O)

May be missing provided an approved procedure is used to ensure that the oxygen supply is at or above minimum requirements for the flight.

Minimum Equipment List Procedures (MELP)

Placard

Inoperative Flight Deck Fixed Oxygen System Overboard Discharge Indicator must be placarded in the flight compartment.

Operating Procedures

1. Prior to each flight, check the pressure in the Flight Deck Fixed Oxygen System bottle by whichever pressure gauge is operative.
2. Verify pressure in oxygen cylinder meets minimum dispatch pressure. Refer to Figure 35-1 and Figure 35-2 for pressure variations with change in ambient temperature.

Maintenance Procedures

1. Placard Flight Deck Fixed Oxygen System Overboard Discharge Indicator on OXYGEN Pressure indicator, Co-pilot's side console panel.
2. Make appropriate entry in the aircraft journey log.

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35-30-1 Protective Breathing Equipment (PBE)

Interval	Installed	Required	Procedure
	4	4	NIL

NOTE: Required distribution is Qty 1 in Flight Deck and Qty 3 in Passenger Cabin.

Minimum Equipment List Procedures (MELP)

Placard

None required.

Operating Procedures

None required.

Maintenance Procedures

None required.

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35-30-2 First Aid Oxygen System

Interval	Installed	Required	Procedure
C	2	1	(M)(O)

As per Company Procedure one may be unserviceable or missing provided:

a) Required distribution of operative units is maintained throughout the aircraft, and
b) Bottles not properly serviced are replaced, serviced, or removed at the next available maintenance facility.

NOTE:

1. Required Distribution is Qty 1 in Passenger Cabin either in AFT or FWD Bulkhead.
2. Applicable only for portable oxygen bottles with two ports.

Minimum Equipment List Procedures (MELP)

Placard

Placard Portable First Aid Oxygen System in the passenger compartment.

Operating Procedures

1. Required distributions of operative units are maintained throughout the aircraft.
2. Cabin crew to conduct the pre-flight check of the equipments and inform the PIC of any defect/loss of pressure in the bottle. The PIC to check the table (Ref page 35-30-3. 3), and ensure that the oxygen quantity is sufficient for the flight. (Ref SEP manual Ch 5, Para 5.1).

Maintenance Procedures

1. To be developed by the operator such that relevant safety standards and regulations are met.
2. Ensure inoperative bottles are removed from their installed locations and secured out of sight or removed from the aircraft prior to dispatch. Bottles being secured out of sight should be stowed away such that they are out of immediate access to either crew or passengers within the aircraft.
3. Placard removed/missing bottles' normal storage location(s), and inoperative first aid oxygen bottles are placarded, if stowed away on the aircraft.
4. Make appropriate entry in the aircraft journey log.

35-30-3 Passenger Portable Oxygen Dispensing Units

35-30-3-1

Interval	Installed	Required	Procedure
C	3	2	NIL

Number of passengers to be restricted up to 58 (for aircraft with 78 seat configuration) or 60 (for aircraft with 90 seat configuration, also find note 2).

NOTE:

1. Above requirements have been tabulated based on following:

"Minimum qty and distribution required for dispatch will depend on the number of passenger onboard. The three cylinders supply 10% of the passenger and sufficient oxygen for at least 30 minutes at cabin pressure altitudes above 10,000 ft. Each cylinder has three masks attached to it catering for a total of 9 Passengers."

2. In case of aircraft with 90 seat configuration, Flight attendant can use additional flight attendant portable oxygen bottle (of 4.3 cu.ft.).
3. Applicable only for portable oxygen bottles with three ports.

Minimum Equipment List Procedures (MELP)

Placard

Placard inoperative Passenger Portable Oxygen Dispensing Units.

Operating Procedures

1. Required distribution of operative unit is maintained throughout the aircraft.
2. Cabin crew to conduct the pre-flight check of the equipments and inform the PIC of any defect/loss of pressure in the bottle. The PIC to check the table (Ref page 35-30-3. 3), and ensure that the oxygen quantity is sufficient for the flight. (Ref SEP manual Ch 5, Para 5.1).

Maintenance Procedures

1. Check minimum dispatch pressure of remaining Passenger Portable Oxygen Dispensing Units. The following table is included which gives minimum dispatch pressure variation with ambient temperature.
2. Ensure that inoperative bottles are removed from their installed locations and secured out of sight or removed from the aircraft prior to dispatch. Bottles being secured out of sight should be stowed away such that they are out of immediate access to either crew or passengers within the aircraft.
3. Placard removed/missing bottles' normal storage location(s), and inoperative Passenger Portable Oxygen System bottles are placarded, if stowed away on the aircraft.
4. Make appropriate entry in the aircraft journey log.

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35-30-3 Passenger Portable Oxygen Dispensing Units

35-30-3-2

Interval	Installed	Required	Procedure
C	3	1	NIL

Number of passengers to be restricted up to 28 (for aircraft with 78 seat configuration) or 30 (for aircraft with 90 seat configuration, also find note 2).

NOTE:

1. Above requirements have been tabulated based on following:

   "Minimum qty and distribution required for dispatch will depend on the number of passenger onboard. The three cylinders supply 10% of the passenger and sufficient oxygen for at least 30 minutes at cabin pressure altitudes above 10,000 ft. Each cylinder has three masks attached to it catering for a total of 9 Passengers."
2. In case of aircraft with 90 seat configuration, Flight attendant can use additional flight attendant portable oxygen bottle (of 4.3 cu.ft.).
3. Applicable only for portable oxygen bottles with three ports.

Minimum Equipment List Procedures (MELP)

Placard

Placard inoperative Passenger Portable Oxygen Dispensing Units.

Operating Procedures

1. Required distribution of operative unit is maintained throughout the aircraft.
2. Cabin crew to conduct the pre-flight check of the equipments and inform the PIC of any defect/loss of pressure in the bottle. The PIC to check the table (Ref page 35-30-3. 3), and ensure that the oxygen quantity is sufficient for the flight. (Ref SEP manual Ch 5, Para 5.1).

Maintenance Procedures

1. Check minimum dispatch pressure of remaining Passenger Portable Oxygen Dispensing Units. The following table is included which gives minimum dispatch pressure variation with ambient temperature.
2. Ensure that inoperative bottles are removed from their installed locations and secured out of sight or removed from the aircraft prior to dispatch. Bottles being secured out of sight should be stowed away such that they are out of immediate access to either crew or passengers within the aircraft.
3. Placard removed/missing bottles' normal storage location(s), and inoperative Passenger Portable Oxygen System bottles are placarded, if stowed away on the aircraft.
4. Make appropriate entry in the aircraft journey log.

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35-30-3 Passenger Portable Oxygen Dispensing Units

PRESSURE VARIATIONS WITH TEMPERATURE

TEMPERATURE (°F)	MINIMUM DISPATCH PRESSURE (PSIG)	MAXIMUM CHARGE PRESSURE (PSIG)
–65	1189	1375
–60	1204	1393
–55	1219	1410
–50	1234	1428
–45	1250	1445
–40	1265	1463
–35	1280	1481
–30	1295	1498
–25	1311	1516
–20	1326	1533
–15	1341	1551
–10	1356	1569
–5	1372	1586
0	1387	1604
5	1402	1621
10	1417	1639
15	1432	1656
20	1448	1674
25	1463	1692
30	1478	1709
35	1493	1727
40	1509	1744
45	1524	1762

(…Continued)

35-30-3 Passenger Portable Oxygen Dispensing Units

PRESSURE VARIATIONS WITH TEMPERATURE (CONT'D)

TEMP °F	MINIMUM DISPATCH	MAXIMUM CHARGE
50	1539	1780
55	1554	1797
60	1570	1815
65	1585	1832
70	1600	1850
75	1615	1868
80	1630	1885
85	1646	1903
90	1661	1920
95	1676	1938
100	1691	1956
105	1707	1973
110	1722	1991
115	1737	2008
120	1752	2026
125	1768	2044
130	1783	2061
135	1798	2079
140	1813	2096
145	1828	2114
150	1844	2131
155	1859	2149
160	1874	2167
165	1889	2184

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